

Training and support for dementia caregivers: towards sustainable, effective long-term care (LTC) system in China

Introduction

The Chinese government's ambitious plan to build a LTC system for its aging population has fueled investment in care homes across the country, resulting in a thriving industry.¹ Although it is widely perceived that Chinese care homes are falling short, the care homes in China have a much lower occupancy rate (50% in 2020²) than over 80% in the developed economies including the UK,³ the US,⁴ and Japan.⁵ There exist mismatched needs and supplies for institutional care. For example, in 2019, only 38.8% of rural care home placements in central China's Anhui province are occupied, while care homes in large cities are out of demand.⁶ Similar mismatch in Beijing has triggered local government to move those in urban care homes to rural ones to balance the needs.⁷ Despite the increasing number of care homes, there is a lack of skilled or trained professional carers to provide quality dementia care, which often makes them unable to accommodate PwDs. Thus, there is an urgent need for the government to review its investment in care homes to reduce resource misallocation and to invest more on improving care capability and quality, rather than the number of care homes alone.

A sustainable, effective LTC system should prioritise the majority of people with dementia (PwD), who are the main source of care needs, as well as their caregivers. The majority of care for PwDs in China is provided by family members (84.9%), with 8.3% living alone, 4.9% supported by paid caregivers, and a mere 2% receiving institutional care.⁸ According to the Chinese government, the Chinese LTC system will be where "home-based care is the foundation, community-based care provides the necessary support, and residential care is supplementary."⁹ Thus, the government should take measures to provide training and support for home- and community-based carers, which is still absent.¹⁰ This means that although institutional care is expected to accommodate more PwDs in the future, changing social acceptance of institutional care may take a long time. Better care quality in the future is insufficient to compensate the current care poverty among the elderly, highlighting a need to prioritise training and support for family caregiver, which is neglected in the current elder care policy.¹¹ Therefore, there is an urgent need to introduce caregiver intervention to the country's long-term care system. This essay first conceptualises caregiver burden in Chinese context, which enables discussion on how to introduce caregiver intervention nationwide, including its content, processes, actors, with supporting evidence, plus future considerations.

Contextualising caregiver burnout in China

Dementia is an umbrella term for brain disorders that result in reduced cognitive function.¹² As a leading cause of disabilities affecting 15.07 million people aged 60 or above in China,^{13,14} it encompasses not only cognitive impairments but also behavioural and psychological symptoms (BPSD) that significantly disrupt

daily life, distinguishing it from mild cognitive impairment.¹⁵ People with dementia (PwDs) require caregivers to help manage their daily routines and prevent potential harm.^{15,16} Due to the complex, diverse aetiologies, it is hard to develop medicine for dementia and even if there is any drug, it is likely to benefit only limited proportion of PwDs, highlighting the importance of long-term care (LTC).^{17,18} Caregivers frequently struggle to balance their caregiving duties with other life responsibilities, such as work and personal relationships, making them the hidden secondary patients of dementia. This struggle typically leads to increased stress, depression, anxiety, and other health problems among caregivers.¹⁹ When caregivers are unable to meet the care demand, the worsening of the disease will increase caregiver burden and reduce their willingness and capability to care, leading to a burden-burnout feedback loop.²⁰

In China, over 80% of PwD are cared by their family, which lead to mental and physical burdens. Caregiver burden is associated with a wider social context of the country (Figure 1), which is different from that in the West. Caregiver burden in China is higher than that in the West due to a lack of community care.²¹ The high level of caregiver burden is prevailing and is associated with depression and anxiety and affected by care dependency.²² A cross-sectional study in Shanghai shows that at least half of community-dwelling PwDs have BPSD, which is associated with lower daily living activities (ADL) scores, suggesting an increasing care dependency.²³ With disease progression, PwDs develop BPSD, such as depression, anxiety, apathy, and agitation. Irritability, agitation, disinhibition, delusions, and hallucinations will eventually be present in severe cases.²⁴ Psychosis, characterised by visual hallucinations without complex delusions and thought disorder, is among the most disruptive BPSD and occurs in 30-50% of AD patients.²⁵ These BPSD often require new skills and treatments, while the failure to meet the demands with sufficient diagnosis and effective treatment may further worsen the symptoms that increases care dependency, leading to a second vicious cycle due to underdiagnosis and undertreatment of PwDs.

Underlying underdiagnosis and undertreatment is poor knowledge and associated stigmatisation. A 2015 survey conducted in five Chinese cities show that 76.6% of people fear Alzheimer's disease (AD), which is associated with their lack of knowledge about AD, as only 10.4% of the people say that they have sufficient AD knowledge.²⁶ People often consider dementia and cognitive decline a natural process of aging, thus unwilling to get diagnosis or treatment. As a result, 90% of dementia cases are never diagnosed, which is associated with low socioeconomic levels, including poor education, low occupational class, low income and living in a rural area.²⁷⁻²⁹ These contribute to a self-strengthening impact of knowledge gap and stigmatisation.

Healthcare providers in China also do not have sufficient knowledge for diagnosis and treatment, which leads to nihilism in diagnosis and treatment.³⁰ For caregivers, when they view dementia as a terminal illness, they will suffer from despair and hopelessness, thus stopping seeking any medicine or interventions to improve care and even wishing the PwD to die.³⁰ The lack of knowledge in dementia care also leads to poor care capability, which makes caregivers unable to meet the growing care responsibility

with the worsening of the disease. People often mis-label the behaviour of PwDs as BPSD, ignoring the unmet needs of PwDs and resulting in further stigmatisation of dementia.²⁴ While feeling lonely and unsupported themselves, caregivers are often reluctant to take PwD they care to public spaces, as they feel ashamed of the behaviour of PwDs.³⁰ Despite the filial piety law to promote care responsibility of adult children and respect for the elderly from society, there is no law to prohibit discrimination,³¹ which instead sustains caregiver burnout.

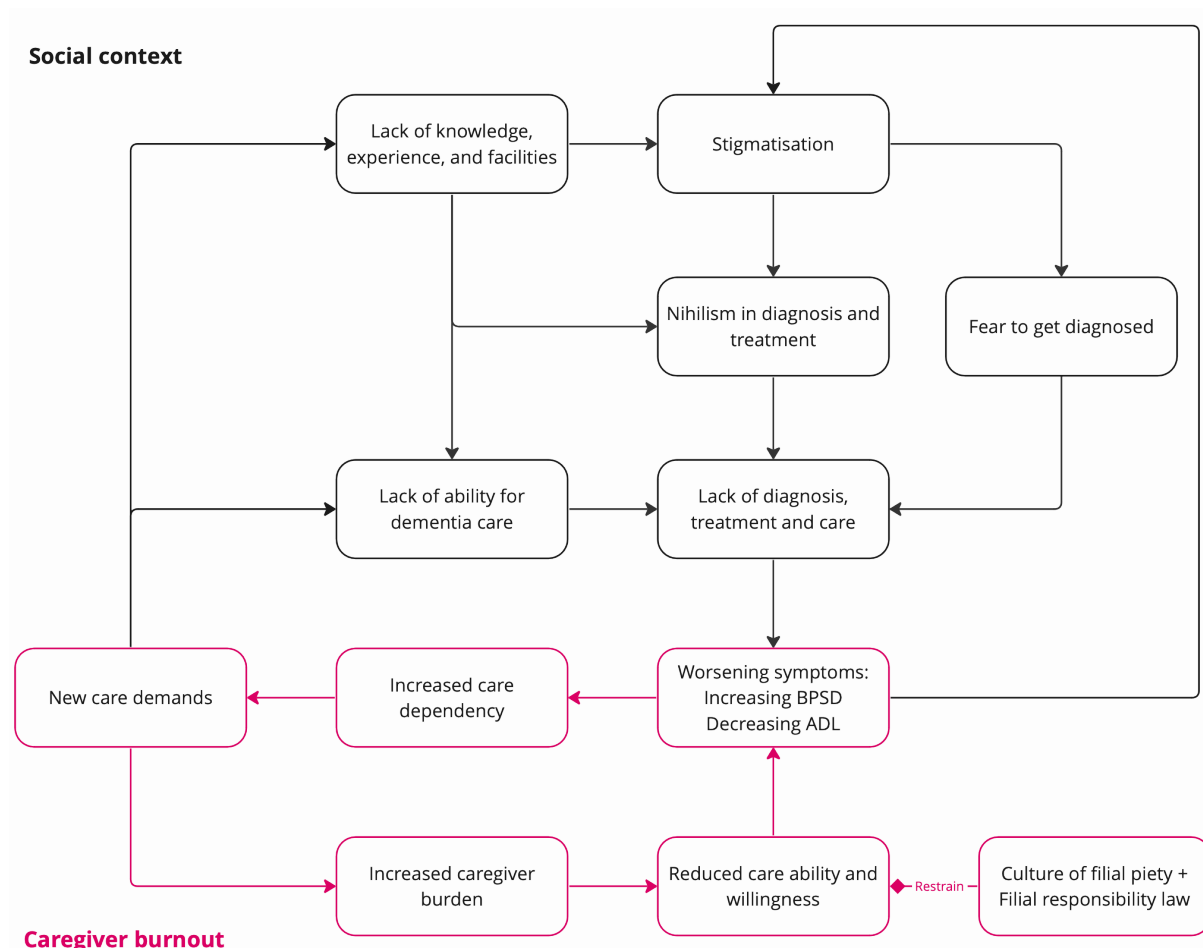


Figure 1. A logical model of caregiver burnout in Chinese context

Implementing caregiver training and support: from evidence to content

The vicious loops in Chinese dementia care presents unique challenges for family caregivers as compared with their Western counterparts (Figure 1). In the West, caregiver intervention is mostly used to alleviate mental distress, while in China, caregivers experience more physical and time-dependence burdens, which is much higher than emotional burden.¹⁰ Thus, targeting caregiver mental health conditions alone will not be sufficient and sustainable due the self-strengthening feedback loops in this context that continuously worsens caregiver burden. Therefore, this essay proposes a three-step approach, which should be included as the core content of the intervention.

Step 1: Ending dementia stigma

Due to the lack of post-diagnosis support for dementia care, diagnosing and treating dementia is unfavourable, as it adds to the chance of being discriminated against. Thus, the starting point of caregiver support in China is to eliminate stigmatisation around dementia, especially the nihilism of diagnosis and treatment by facilitating the right knowledge and attitude to dementia diagnosis and treatment, or else PwDs and their caregivers may refuse to get diagnosed and receive any treatment and intervention,³¹ which will undermine any effort to provide caregiver interventions, as they will feel such interventions meaningless and even leads to abuse.³² Also, the Chinese terminology for dementia means “insane and idiotic”, which has a negative connotation contributing to stigmatisation, making people with dementia-like symptoms reluctant to get a diagnosis.³³ Thus, there is a need to disseminate dementia knowledge through public media and government channels,³⁴ of which Taiwan and Japan have set an example for talking about dementia in appropriate language, excluding stigmatised phrases.³³ When implementing caregiver intervention at a community level, it is important to first discuss about dementia stigmatisation to eliminate nihilism of diagnosis and treatment and stigmatisation about BPSD, ensuring caregivers to understand why PwDs need diagnosis and treatment.^{35,36}

Step 2: Coping skills for BPSD

In China, home care does not mean that PwD have a right to be home as they like. Often, home is a prison to the elderly,³⁷ as their family fears PwD of appearing ‘weird’ or getting lost outside.³⁰ The fear for spatial disorientation and other BPSD is likely due to their lack of knowledge and coping skills for the BPSD, which contribute to dementia-related stigma and caregiver burden, especially physical burden, as PwD, regarded a constant risk, require care all the time at home. This psychosocial backgrounds contribute to a different patterns of BPSD.³⁸ For example, sleep disorder is one of the most common BPSD, which leads to caregiver's vigilance at night-time. Continued sleep disturbances in PwD may worsen caregiver burden.³⁸ The differences in BPSD patterns also leads to different care experiences. Compared with their western counterparts, BPSD severity is more strongly associated with caregiver burden in China.¹⁰ Thus, when providing caregiver intervention to Chinese caregivers, the coping strategies to the unique BPSD patterns in China must be delivered.³⁹ Also, caregivers should be taught to communicate, identify and satisfy the unmet needs of PwD underlying their behaviour.^{24,35,40,41} The knowledge about BPSD, especially how to tell BPSD from normal behaviour and how to address them, will enable more autonomy for PwD, which is expected to alleviate physical burden for caregivers.

Step 3: Self-care and service user leadership

Wu et al (2018) have identified 19 effective interventions to alleviate caregiver burden in Chinese caregivers,¹¹ with Shi et al (2020) expected to explore more relevant interventions.⁴² The typical

components of these interventions include behavioural management, psychosocial support, dementia education, case management and coping strategies for caregiver distress. However, as Zarit (2018) argues, some caregivers seek help not because of they are depressed and anxious, while most interventions are designed to improve caregiver distress,⁴³ indicating mismatched expectations. Since most interventions targeting caregiver distress, regardless of its methods and content, have been found to be effective, it is critical understand what caregivers want the most from intervention.¹¹ Thus, service user and survivor's experience should be placed in the central role in choosing and designing caregiver intervention components.^{44,45} Especially, social support within the service user community is critical to information dissemination, where a WeChat-based group training is a promising approach to share critical knowledge about dementia care and self-care among caregivers.⁴⁶

Ensuring accessibility and equality: challenges for delivery

Where to intervene: community service as a hub

To make caregiver interventions accessible to all, trained carers need to be deployed at a community level. Community services are the foundation for any initiative aimed at improving the quality of informal care. In addition to imparting crucial knowledge about dementia care to family caregivers, community-based professional carers also reach out to elderly individuals living alone, who are less likely to consult a doctor, thus ensuring health equity by addressing the underutilisation of healthcare resources.^{47,48} The focus on community-based care aligns with a broader global trend, as high-income countries increasingly prioritise community-based care over institutional care.⁴⁹ Hu (2019) demonstrated that community care can significantly alleviate the pressure on formal care.⁹ These programmes should also emphasize the importance of regular check-ups and medication adherence, as well as the benefits of healthy lifestyle choices such as exercise and a balanced diet, especially after hospital discharge.⁵⁰

Who to intervene: training local workforce

However, expanding the professional nursing workforce is constrained by educational limitations and can be both time-consuming and costly, which makes it unlikely to be the main source of interventionists in the short-term. The 10/66 intervention trials in India, Peru, and Russia, which train community workers as home-care advisors, may offer a cost-effective solution to address workforce shortages^{51–54}. Thus, policymakers should consider leveraging and expanding the existing community services to provide more accessible and culturally appropriate caregiver support. Notably, as government reforms have produced redundant public services team well-represented in local communities, who needs to be re-tasked, Sylvia (2021) has demonstrated with a randomised controlled trial that the cadres from the former Family Planning Committee were able to re-tasked for delivering effective parenting interventions⁵⁵. Whether re-

training and re-tasking of the local cadres may need further research, but this may alleviate the current nursing workforce shortage.

How to intervene: considerations for cultural appropriateness

A challenge for community-based delivery of caregiver intervention is the cultural appropriateness of the programme. As younger generations in China become more urbanised, educated, and exposed to diverse social values, older adults may feel increasingly marginalised when facing the socioeconomic evolution.⁵⁶ In the age where the elderly people grew up, people lack the knowledge for public health and do not understand how they can benefit from such programmes.^{57,58} Low literacy is common among the females and the elderly, which makes them unable to access and understand health information.⁵⁹ A further barrier to effective intervention is the language, as the younger generations and health professionals tend to speak Mandarin, while most elderly people speak a certain Chinese dialect.^{60,61} These factors contribute to low health literacy among the elderly, which needs to be addressed in effective caregiver intervention.⁶² Therefore, any caregiver intervention should be made available with easy-to-understand content tailored to local languages, not limited to Mandarin, to improve its accessibility. This again stresses the importance of training local workforce and needs to be addressed at a community level.

Towards sustainability and resilience: focusing on long-term impact

Political leadership

In China, healthcare and eldercare are under the National Health Commission (NHC) and the Ministry of Civil Affairs (MCA), respectively, whose services are both paid by the Ministry of Human Resources and Social Security (MHRSS), which is coordinated through the multi-ministerial National Working Commission on Ageing (NWCA).^{63,64} The recent government restructuring has announced that MCA will take the lead of lead NWCA from NHC for more effective elder care policy making and delivery.⁶⁵ Thus, to implement caregiver intervention, it is critical to obtain support from NWCA, especially MCA. NHC and MCA has begun to offer community-based care in the Healthy Eldercare Pilot Demonstration Communities (HEPDC) programmes across China, in which caregiver intervention can be implemented.⁶⁶ China has a long, successful history of policy experimentation, which means small-scale trials before rolling out nationwide, which shows positive selection of effective policy and facilitates policy learning.⁶⁷ Thus, implementing caregiver intervention within HEPDC may be a critical first step before being rolled out nationally.

Sustainable financing

To obtain sustainable finance for caregiver intervention, it is critical to obtain further evidence about the cost-effectiveness of caregiver interventions, to inform policymakers of the benefits and returns of implementing caregiver intervention, because the benefits of caregiver intervention may not be very

immediate and visible to policymakers. Long-term care insurance (LTCI) could be a source of financing, which has been experimented in 15 Chinese cities since 2016.⁶⁸ LTCI has proven to improve regional and income inequalities in healthcare, thus improving affordability and equitability of healthcare.⁶⁹ A recent systematic review on the impact of LTCI in China has demonstrated its positive effects on various outcomes.⁷⁰ Thus, LTCI is likely to be the main source of finance for caregiver intervention, when it is gradually being rolled out nationwide. Another potential financial support to the intervention is the property management entities. In China, local residents are typically organised by their resident community, which is responsible for coordinating group activities and negotiating with real estate developers and property management companies to maintain life quality and house prices within the community.⁷¹ Thus, caregiver intervention could be introduced as a community activity, especially in an aging community, during which the government should provide a flexible payment structure and resources to meet the demand. For example, the local government could form a dedicated team to help deliver partially- or fully-funded intervention in different communities.

Reforming nursing education

In the long run, China must reform its nursing education system and take measures to alleviate its nursing shortage.⁷² Rodgers et al. (2015)⁷³ noted that although nursing is not considered an ideal job, most educational institutions can still recruit enough students through the current education system. Therefore, an education reform to recruit more students into nursing is needed to address the nursing workforce shortage. For example, Yun et al (2010)⁷² proposes that cooperation with the military medical schools and recruiting retired corpsmen could be a feasible policy options to enlarge the nursing workforce, as military medical schools have high-quality nursing education. Furthermore, more knowledge and training for dementia care, especially how to deliver caregiver interventions in community settings, should be included in nursing education, to make the education more practical. As low-quality care provided in the community health centres has driven local residents to seek help in larger hospitals, despite the travel costs,⁷⁴ it is critical to support life-long learning in primary care settings to improve care quality at the community level. This means that community health centres must have its own access to latest guidelines and medical knowledge about dementia. A solution is to facilitate cooperation and pairing between local communities and hospitals, which has been experimented in Shanghai.⁷⁵ However, for wider regions, the Chinese government should consider establishing or funding a dedicated organisation to provide continuous online and onsite training for community-based nurses, who will then be able to deliver updated medical knowledge to caregivers at the community level.

Designing better caregiver interventions

Designing and testing caregiver interventions with larger scale and better quality will provide further evidence for implementing caregiver interventions in China. To date, most randomised clinical trials about

dementia care have been conducted in high-income countries. One notable intervention is the Resources for Enhancing Alzheimer's Caregiver Health (REACH) trials funded by the National Institutes of Health,^{76,77} the largest trials on caregiver interventions which have been widely adapted in high-income settings.^{78–80} Thus, it is crucial for China to learn from existing trials and gather evidence tailored to its context.⁸¹ Additional attention should be paid to the regional disparities, especially between rural and urban areas. Most of the existing trials for caregiver intervention take place in Hong Kong and Taiwan and highly developed cities in mainland China, which are often very different in designs and have lower quality and smaller scale to Western ones.^{11,42} Thus, there is a need to develop China's REACH trials as a model caregiver intervention, to better address the unique challenges of Chinese family caregivers.

Conclusion

Addressing the growing dementia care challenge in China requires a multifaceted approach that encompasses ending dementia stigma, coping skills for BPSD and ensuring service user leadership, which needs to be implemented in a community level. Thus, it is critical to train local workforce and deliver the intervention in a culturally appropriate way. To make the system sustainable and resilient, it is important to coordinate major political actors within NWCA, achieving financial sustainability with LTIC and other measures, reforming nursing education to train more professional nurses and provide more continuous education opportunities for them. Ultimately, designing culturally appropriate and well-adapted caregiver interventions that consider the specific needs of Chinese caregivers, with more rigorous approaches to test it on larger population, will play a vital role in building a sustainable, resilient LTC in China.

References

1. Feng Z, Zhan HJ, Feng X, Liu C, Sun M, Mor V. An Industry in the Making: The Emergence of Institutional Elder Care in Urban China. *J Am Geriatr Soc*. 2011;59(4):738-744. doi:10.1111/j.1532-5415.2011.03330.x
2. Feng Y. Increasing number of Chinese to rely on nursing homes in old age. CGTN. Published May 12, 2021. Accessed April 13, 2023. <https://news.cgtn.com/news/2021-05-12/Increasing-number-of-Chinese-to-rely-on-nursing-homes-in-old-age-10d10aluLni/index.html>
3. Knight Frank. Care home occupancy rate in the United Kingdom (UK), from 2006 to 2021. Statista. Published November 30, 2021. Accessed April 13, 2023. <https://www.statista.com/statistics/1231777/care-home-occupancy-in-the-uk/>
4. Kaiser Family Foundation. Occupancy rate of certified nursing facilities in the United States, from 2003 to 2022. Statista. Published October 10, 2022. Accessed April 13, 2023. <https://www.statista.com/statistics/1223881/occupancy-rate-of-certified-nursing-facilities-in-the-united-states/>
5. Ministry of Health, Labour and Welfare of Japan. Usage rate of nursing homes for the elderly per facility in Japan in 2021, by institution type. Statista. Published December 27, 2022. Accessed April 13, 2023. <https://www.statista.com/statistics/1211834/japan-occupancy-rate-nursing-homes-by-type/>
6. Fan S, Tai S, Pan Q, et al. [Supply and demand imbalance in elderly care: “difficult to secure a bed” coexists with idle resources]. *Economic Information Daily*. http://www.jjckb.cn/2019-03/25/c_137921075.htm. Published June 11, 2021. Accessed April 13, 2023.
7. Liu M. Empty rural care homes get bed boost. *Global Times*. Published October 23, 2012. Accessed April 13, 2023. <https://www.globaltimes.cn/content/739836.shtml>
8. Jia J, Zuo X, Jia XF, et al. Diagnosis and treatment of dementia in neurology outpatient departments of general hospitals in China. *Alzheimers Dement*. 2016;12(4):446-453. doi:10.1016/j.jalz.2015.06.1892
9. Hu B 1 1 P, Political Science L. Projecting future demand for informal care among older people in China: the road towards a sustainable long-term care system. Published online January 2019:61-81. doi:10.1017/S1744133118000221

10. Wang J, Xiao LD, He GP, Ullah S, De Bellis A. Factors contributing to caregiver burden in dementia in a country without formal caregiver support. *Aging Ment Health*. 2014;18(8):986-996. doi:10.1080/13607863.2014.899976
11. Wu B, Petrovsky DV, Wang J, et al. Dementia caregiver interventions in Chinese people: A systematic review. *J Adv Nurs*. 2019;75(3):528-542. doi:10.1111/jan.13865
12. Cloak N, Al Khalili Y. Behavioral And Psychological Symptoms In Dementia. In: *StatPearls*. StatPearls Publishing; 2023. Accessed April 10, 2023. <http://www.ncbi.nlm.nih.gov/books/NBK551552/>
13. Feigin VL, Nichols E, Alam T, et al. Global, regional, and national burden of neurological disorders, 1990–2016: a systematic analysis for the Global Burden of Disease Study 2016. *Lancet Neurol*. 2019;18(5):459-480. doi:10.1016/S1474-4422(18)30499-X
14. Ren R, Qi J, Lin S, et al. The China Alzheimer Report 2022. *Gen Psychiatry*. 2022;35(1):e100751. doi:10.1136/gpsych-2022-100751
15. Knopman DS, Petersen RC. Mild Cognitive Impairment and Mild Dementia: A Clinical Perspective. *Mayo Clin Proc*. 2014;89(10):1452-1459. doi:10.1016/j.mayocp.2014.06.019
16. Kim B, Noh GO, Kim K. Behavioural and psychological symptoms of dementia in patients with Alzheimer's disease and family caregiver burden: a path analysis. *BMC Geriatr*. 2021;21(1):160. doi:10.1186/s12877-021-02109-w
17. Kales HC. Seeing the forest but missing some important trees. *Lancet Healthy Longev*. 2022;3(4):e225-e226. doi:10.1016/S2666-7568(22)00077-0
18. Wong G, Knapp M. Should we move dementia research funding from a cure to its care? *Expert Rev Neurother*. 2020;20(4):303-305. doi:10.1080/14737175.2020.1735364
19. Brodaty H, Donkin M. Family caregivers of people with dementia. *Dialogues Clin Neurosci*. 2009;11(2):217-228. doi:10.31887/dcns.2009.11.2/hbrodaty
20. Gérard P, Zech E. Informal Caregiver Burnout? Development of a Theoretical Framework to Understand the Impact of Caregiving. *Front Psychol*. 2019;10:1748. doi:10/ggwhgw
21. Zhao M, Zhu Z, Kong C, Zhao C. Caregiver burden and parenting stress among left-behind elderly individuals in rural China: a cross-sectional study. *BMC Public Health*. 2021;21(1):846. doi:10.1186/s12889-021-10892-9

22. Lou Q, Liu S, Huo YR, Liu M, Liu S, Ji Y. Comprehensive analysis of patient and caregiver predictors for caregiver burden, anxiety and depression in Alzheimer's disease. *J Clin Nurs*. 2015;24(17-18):2668-2678. doi:10.1111/jocn.12870
23. Haibo X, Shifu X, Pin NT, et al. Prevalence and severity of behavioral and psychological symptoms of dementia (BPSD) in community dwelling Chinese: findings from the Shanghai three districts study. *Aging Ment Health*. 2013;17(6):748-752. doi:10.1080/13607863.2013.781116
24. Warren A. Behavioral and Psychological Symptoms of Dementia as a Means of Communication: Considerations for Reducing Stigma and Promoting Person-Centered Care. *Front Psychol*. 2022;13:875246. doi:10.3389/fpsyg.2022.875246
25. Agüera-Ortiz L, Babulal GM, Bruneau MA, et al. Psychosis as a Treatment Target in Dementia: A Roadmap for Designing Interventions. *J Alzheimers Dis*. 1970;88(4):1203-1228. doi:10.3233/JAD-215483
26. Zeng F, Xie WT, Wang YJ, et al. General Public Perceptions and Attitudes toward Alzheimer's Disease from Five Cities in China. *J Alzheimers Dis*. 2015;43(2):511-518. doi:10.3233/JAD-141371
27. Lang L, Clifford A, Wei L, et al. Prevalence and determinants of undetected dementia in the community: a systematic literature review and a meta-analysis. *BMJ Open*. 2017;7(2):e011146. doi:10.1136/bmjopen-2016-011146
28. Chen R, Hu Z, Chen RL, Ma Y, Zhang D, Wilson K. Determinants for undetected dementia and late-life depression. *Br J Psychiatry*. 2013;203(3):203-208. doi:10.1192/bjp.bp.112.119354
29. Chen R, Ma Y, Wilson K, et al. A multicentre community-based study of dementia cases and subcases in older people in China—the GMS-AGECAT prevalence and socio-economic correlates. *Int J Geriatr Psychiatry*. 2012;27(7):692-702. doi:10.1002/gps.2767
30. Zhang X, Clarke CL, Rhynas SJ. Tensions in dementia care in China: An interpretative phenomenological study from Shandong province. *Int J Older People Nurs*. 2020;15(1):e12291. doi:10.1111/opn.12291
31. Jia J, Ning Y, Chen M, Wang S, Li Y, Yang H. Ending age discrimination and stigma to promote healthy ageing in China. *The Lancet*. 2022;400(10367):1907-1909. doi:10.1016/S0140-6736(22)02362-5
32. Fang B, Yan E. Abuse of Older Persons With Dementia: A Review of the Literature. *Trauma Violence Abuse*. 2018;19(2):127-147. doi:10.1177/1524838016650185

33. Chiu HFK, Sato M, Kua EH, et al. Renaming dementia – an East Asian perspective. *Int Psychogeriatr*. 2014;26(6):885-887. doi:10.1017/S1041610214000453
34. Hsiao HY, Liu Z, Xu L, Huang Y, Chi I. Knowledge, Attitudes, and Clinical Practices for Patients With Dementia Among Mental Health Providers in China: City and Town Differences. *Gerontol Geriatr Educ*. 2016;37(4):342-358. doi:10.1080/02701960.2014.990152
35. Wolverson E, Birtles H, Moniz-Cook E, James I, Brooker D, Duffy F. Naming and Framing the Behavioural and Psychological Symptoms of Dementia (BPSD) Paradigm: Professional Stakeholder Perspectives. *OBM Geriatr*. 2019;3(4):1-25. doi:10.21926/obm.geriatr.1904080
36. Volkow ND, Gordon JA, Koob GF. Choosing appropriate language to reduce the stigma around mental illness and substance use disorders. *Neuropsychopharmacology*. 2021;46(13):2230-2232. doi:10.1038/s41386-021-01069-4
37. Zhang X, Clarke CL, Rhynas SJ. A thematic analysis of Chinese people with dementia and family caregivers' experiences of home care in China. *Dementia*. 2020;19(8):2821-2835. doi:10.1177/1471301219861466
38. Liu S, Li C, Shi Z, et al. Caregiver burden and prevalence of depression, anxiety and sleep disturbances in Alzheimer's disease caregivers in China. *J Clin Nurs*. 2017;26(9-10):1291-1300. doi:10.1111/jocn.13601
39. Yu H, Wu L, Chen S, Wu Q, Yang Y, Edwards H. Caregiving burden and gain among adult-child caregivers caring for parents with dementia in China: the partial mediating role of reciprocal filial piety. *Int Psychogeriatr*. 2016;28(11):1845-1855. doi:10.1017/S1041610216000685
40. Caspi E. Time for Change: Persons With Dementia and "Behavioral Expressions," Not "Behavior Symptoms." *J Am Med Dir Assoc*. 2013;14(10):768-769. doi:10.1016/j.jamda.2013.05.019
41. Pot AM, Oliveira D, Hoffman J. Towards healthy ageing in China: shaping person-centred long-term care. *The Lancet*. 2022;400(10367):1905-1906. doi:10.1016/S0140-6736(22)02361-3
42. Shi C, Chen S, Salcher-Konrad M, et al. Effectiveness of interventions for people living with dementia and their carers in Chinese communities: protocol for a systematic review and meta-analysis of randomised controlled trials. *BMJ Open*. 2021;11(8):e047560. doi:10.1136/bmjopen-2020-047560
43. Zarit SH. Past is prologue: how to advance caregiver interventions. *Aging Ment Health*. 2018;22(6):717-722. doi:10.1080/13607863.2017.1328482

44. Faulkner A, Carr S, Gould D, et al. 'Dignity and respect': An example of service user leadership and co-production in mental health research. *Health Expect*. 2021;24(S1):10-19. doi:10.1111/hex.12963
45. Callard F, Rose D. The mental health strategy for Europe: Why service user leadership in research is indispensable. *J Ment Health*. 2012;21(3):219-226. doi:10.3109/09638237.2011.651661
46. Wang F, Xiao LD, Wang K, Li M, Yang Y. Evaluation of a WeChat-based dementia-specific training program for nurses in primary care settings: A randomized controlled trial. *Appl Nurs Res*. 2017;38:51-59. doi:10.1016/j.apnr.2017.09.008
47. Liu LJ, Sun X, Zhang CL, Guo Q. Health-care Utilization Among Empty-Nesters in the Rural Area of a Mountainous County in China. *Public Health Rep*. 2007;122(3):407-413. Accessed April 12, 2023. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1847485/>
48. Zhou C, Ji C, Chu J, et al. Non-use of health care service among empty-nest elderly in Shandong, China: a cross-sectional study. *BMC Health Serv Res*. 2015;15(1):294. doi:10.1186/s12913-015-0974-1
49. Knapp M, Comas-Herrera A, Somani A, Banerjee S. Dementia: international comparisons. Published online 2007.
50. Zhao W, Wu ML (Winnie), Petsky H, Moyle W. Family carers' expectations regarding dementia care services and support in China: A qualitative study. *Dementia*. 2022;21(6):2004-2019. doi:10.1177/14713012221106817
51. Gavrilova SI, Ferri CP, Mikhaylova N, Sokolova O, Banerjee S, Prince M. Helping carers to care--the 10/66 dementia research group's randomized control trial of a caregiver intervention in Russia. *Int J Geriatr Psychiatry*. 2009;24(4):347-354. doi:10.1002/gps.2126
52. Guerra M, Ferri CP, Fonseca M, Banerjee S, Prince M. Helping carers to care: the 10/66 dementia research group's randomized control trial of a caregiver intervention in Peru. *Rev Bras Psiquiatr*. 2011;33(1):47-54. doi:10/bmn2ch
53. Dias A, Dewey ME, D'Souza J, et al. The Effectiveness of a Home Care Program for Supporting Caregivers of Persons with Dementia in Developing Countries: A Randomised Controlled Trial from Goa, India. Brayne C, ed. *PLoS ONE*. 2008;3(6):e2333. doi:10.1371/journal.pone.0002333
54. Prina AM, Mayston R, Wu YT, Prince M. A review of the 10/66 dementia research group. *Soc Psychiatry Psychiatr Epidemiol*. 2019;54(1):1-10. doi:10.1007/s00127-018-1626-7

55. Sylvia S, Warrinnier N, Luo R, et al. From Quantity to Quality: Delivering a Home-Based Parenting Intervention Through China's Family Planning Cadres. *Econ J*. 2021;131(635):1365-1400. doi:10.1093/ej/ueaa114
56. Warmenhoven H, Hoebink PRJ, Janssens JMAM. The Chinese Postreform Generation as Caregivers: The Caregiving Intentions Toward Parents and Parents-in-Law of the One-Child Generation. *J Fam Issues*. 2018;39(14):3690-3712. doi:10.1177/0192513X18789208
57. Chen S, Boyle LL, Conwell Y, Xiao S, Chiu HFK. The challenges of dementia care in rural China*. *Int Psychogeriatr*. 2014;26(7):1059-1064. doi:10.1017/S1041610214000854
58. Zhao W. Dementia care in China: Challenges and recommendations. *Alzheimers Dement*. 2021;17(S8):e049510. doi:10.1002/alz.049510
59. Lam T, Cheng Y, Chan Y. Low literacy Chinese patients: how are they affected and how do they cope with health matters? A qualitative study. *BMC Public Health*. 2004;4:14. doi:10.1186/1471-2458-4-14
60. Zhang D, Jiang Z, Xie Y, et al. Linguistic barriers and healthcare in China: Chaoshan vs. Mandarin. *BMC Health Serv Res*. 2022;22(1):376. doi:10.1186/s12913-022-07744-6
61. Lu S, Chen S, Wang P. Language barriers and health status of elderly migrants: Micro-evidence from China. *China Econ Rev*. 2019;54:94-112. doi:10.1016/j.chieco.2018.10.011
62. Chen X, Giles J, Yao Y, et al. The path to healthy ageing in China: a Peking University–Lancet Commission. *The Lancet*. 2022;400(10367):1967-2006. doi:10.1016/S0140-6736(22)01546-X
63. Krings MF, van Wijngaarden JDH, Yuan S, Huijsman R. China's Elder Care Policies 1994–2020: A Narrative Document Analysis. *Int J Environ Res Public Health*. 2022;19(10):6141. doi:10.3390/ijerph19106141
64. Nan Y, Feng T, Hu Y, Qi X. Understanding Aging Policies in China: A Bibliometric Analysis of Policy Documents, 1978–2019. *Int J Environ Res Public Health*. 2020;17(16):5956. doi:10.3390/ijerph17165956
65. Times G. China to improve elderly-care work mechanism, expand services for seniors - Global Times. Published March 8, 2023. Accessed May 5, 2023. <https://www.globaltimes.cn/page/202303/1286843.shtml>

66. Li X, Yang A, Yan H. Priorities and Instruments of Local Elderly Care Policies in China: Text Mining and Comparative Analysis. *Front Public Health*. 2021;9. Accessed May 5, 2023. <https://www.frontiersin.org/articles/10.3389/fpubh.2021.647670>
67. Wang S, Yang DY. Policy Experimentation in China: the Political Economy of Policy Learning. Published online October 2021. doi:10.3386/w29402
68. Dai W, Li Y, Shen J. The Pilot Programme of Long-term Care Insurance in China: Fragmentation and Policy Implications. *China Int J*. 2022;20(2):186-206. doi:10.1353/chn.2022.0010
69. Liu H, Hu T. Evaluating the long-term care insurance policy from medical expenses and health security equity perspective: evidence from China. *Arch Public Health*. 2022;80(1):3. doi:10.1186/s13690-021-00761-7
70. Chen S, Li L, Yang J, et al. The impact of long-term care insurance in China on beneficiaries and caregivers: A systematic review. *J Glob Health Econ Policy*. 2021;1:e2021014. doi:10.52872/001c.29559
71. Tang B. Grid Governance in China's Urban Middle-class Neighbourhoods. *China Q*. 2020;241:43-61. doi:10.1017/S0305741019000821
72. Yun H, Jie S, Anli J. Nursing shortage in China: State, causes, and strategy. *Nurs Outlook*. 2010;58(3):122-128. doi:10.1016/j.outlook.2009.12.002
73. Rodgers S, Zhu J, Melia K. Can education resolve nursing shortage in China? *Athens J Health*. 2015;2(3):177-190. Accessed April 12, 2023. <https://www.research.ed.ac.uk/en/publications/can-education-resolve-nursing-shortage-in-china>
74. Bhandari B. China's Community Health Centers Are Losing Public Appeal. SixthTone. Published July 15, 2022. Accessed May 14, 2023. <https://www.sixthtone.com/news/1010777>
75. Yang D liang, Li X, Su N, Ji W dong, Chen X xuan. Family medical intervention model of senile dementia with behavioral and psychological symptoms. *J Shanghai Jiao Tong Univ Med Sci*. 2017;37(3):398-402. doi:10.3969/j.issn.1674-8115.2017.03.024
76. Elliott AF, Burgio LD, DeCoster J. Enhancing Caregiver Health: Findings from the Resources for Enhancing Alzheimer's Caregiver Health II Intervention. *J Am Geriatr Soc*. 2010;58(1):30-37. doi:10.1111/j.1532-5415.2009.02631.x

77. Schulz R, Burgio L, Burns R, et al. Resources for Enhancing Alzheimer's Caregiver Health (REACH): Overview, Site-Specific Outcomes, and Future Directions. *The Gerontologist*. 2003;43(4):514-520. Accessed April 11, 2023. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2579756/>
78. Heinrich S, Berwig M, Simon A, et al. German adaptation of the Resources for Enhancing Alzheimer's Caregiver Health II: study protocol of a single-centred, randomised controlled trial. *BMC Geriatr*. 2014;14(1):21. doi:10.1186/1471-2318-14-21
79. Martindale-Adams J, Tah T, Finke B, LaCounte C, Higgins BJ, Nichols LO. Implementation of the REACH model of dementia caregiver support in American Indian and Alaska Native communities. *Transl Behav Med*. 2017;7(3):427-434. doi:10.1007/s13142-017-0505-1
80. Nichols LO, Martindale-Adams J, Burns R, Graney MJ, Zuber J. Translation of a Dementia Caregiver Support Program in a Health Care System—REACH VA. *Arch Intern Med*. 2011;171(4):353-359. doi:10.1001/archinternmed.2010.548
81. *Beyond Filial Piety: Rethinking Aging and Caregiving in Contemporary East Asian Societies*. Vol 6. 1st ed. Berghahn Books; 2020. doi:10.2307/j.ctv21hrg88